EXECUTIVE BRIEF · SAUDI ARABIA BENCHMARK v6.1 (ALLOW_SYNTHETIC=FALSE)

TerraShift AI: Multi-Modal Desert Change Intelligence

Fusing Siamese U-Net Spatial Priors, Google AlphaEarth V1 (64-D) Embeddings & Sentinel-2 Spectral Physics

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79.2% / 82.5%
HYBRID F1 (POOLED / MACRO)
TerraShift v6.0 Hybrid · 100% Recall

53.5% / 64.1%
5-DAY TACTICAL F1 (V5.1)
Zero-GEE Sentinel-2 L2A Autonomy

-91.2% FPs
DESERT FALSE-ALARM SUPPRESSION
114 → 10 FPs · 0 Control-Zone Hits

91.19 km²
4 VERIFIED SAUDI CORRIDORS
19 Ground-Truth Events · 1.485s CPU

1. EVOLUTION FROM ORIGINAL CONCEPT TO VERIFIED FINAL IMPLEMENTATION (V6.1)

Original Concept vs. Saudi Arid-Zone Reality

- **Original Idea:** Ingest multi-date satellite imagery over coordinates, filter seasonal/lighting variations, and segment/quantify physical ground changes (construction, clearing, coastal reclamation).
- **Arid Failure Mode Discovered:** In hyper-arid Saudi terrains ($\alpha > 0.45$), raw Siamese U-Net produces **114 false alarms** (23.5% Pooled F1), while standalone **Google AlphaEarth V1** triggers **90 false alarms** (27.0% F1) and **Google Dynamic World V1** misses 68.4% of desert grading (19.7% F1).

Final Two-Tier Architecture Implemented (v6.1)

- **Tier 1 — Strategic Annual Verification (`alphaearth_hybrid`, v6.0):** Fuses Siamese U-Net boundaries with authenticated Google AlphaEarth V1 64-D embeddings (`project='terrashift'`) and directional Sentinel-2 L2A spectral guards ($\Delta NDBI \geq +0.025$, $|\Delta NDVI| \geq 0.12$).
- **Tier 2 — 5-Day Tactical Autonomy (`siamese_tuned_filtered`, v5.1):** Operates directly on 5-day Copernicus Sentinel-2 L2A STAC streams without waiting for annual GEE embeddings (53.5% Pooled / 64.1% Macro F1).

2. VERIFIED 7-MODEL SAUDI BENCHMARK LEADERBOARD (19 EVENTS · 91.19 KM² · LOCALIZATION-AWARE 1-TO-1)

MODEL ARCHITECTURE	DATA MODALITY	POOLED P	MACRO P	RECALL	POOLED F1	MACRO F1	FPS	CTRL HITS
1. TerraShift v6.0 Hybrid (`alphaearth_hybrid`)	**S2 6-Band + AlphaEarth 64-D**	**65.5%**	**70.1%**	**100.0%**	**79.2%**	**82.5%**	**10**	**0**
2. TerraShift v5.1 Tuned (`siamese_tuned_filtered`)	**Sentinel-2 L2A 6-Band (5-Day)**	**36.5%**	**48.5%**	**100.0%**	**53.5%**	**64.1%**	**33**	**0**
3. Spectral CVA Baseline (`cva_baseline`)	Sentinel-2 L2A Spectral	23.7%	23.3%	94.7%	37.9%	37.3%	58	13
4. Google AlphaEarth V1 Standalone (`alphaearth_cosine`)	GEE AlphaEarth V1 64-D	15.7%	18.4%	94.7%	27.0%	30.5%	90	23
5. Raw Siamese U-Net (`siamese_raw`)	Sentinel-2 L2A 6-Band	13.3%	15.8%	100.0%	23.5%	26.6%	114	22
6. Multi-Sensor Early Fusion (`early_fusion_baseline`)	S2 + Landsat + SAR	12.9%	12.7%	89.5%	22.5%	22.2%	98	17
7. Google Dynamic World V1 Standalone (`dynamic_world_lulc`)	GEE Dynamic World V1	14.3%	17.3%	31.6%	19.7%	22.3%	36	4

3. 4 VERIFIED CORRIDORS (INCL. HELD-OUT AL-KHARJ) & 7-WORKSPACE WEB PLATFORM (NEXT.JS 16 + FASTAPI)

4 Multi-Biome Saudi Corridors (91.19 km²)

- **Riyadh ROSHN Sedra (17.88 km² · 5 GT):** 83.3% Hybrid F1 vs. 21.3% standalone AlphaEarth.
- **Jeddah North Obhur (18.31 km² · 5 GT):** 90.9% Hybrid F1 on coastal reclamation; zero sabkha FPs.
- **Al-Jouf Wadi As-Sirhan (26.82 km² · 5 GT):** 71.4% Hybrid F1 on center-pivot irrigated agriculture.
- **Al-Kharj Industrial [HELD-OUT] (28.18 km² · 4 GT):** Zero hyperparameter tuning; 84.2% Hybrid F1 vs. 0.0% Dynamic World V1.

7 Production Web Workspaces (1.485s Local CPU)

- **`/` & `/analyze`:** Dual synchronized MapLibre GL canvas, live SSE telemetry, WGS84 geodesic clipping, PDF/GeoJSON export.
- **`/benchmarks` & `/timeline`:** 7-model Saudi leaderboard, Grad-CAM++ XAI heatmaps, BFAST 2017–2024 breakpoints & 2030 contagion forecast.
- **`/lab`, `/carbon`, `/watchlist`:** Custom pair upload, IPCC Tier-1 4-pool carbon & $Delta$LST UHI valuation, live AWS STAC v1.0.0 discovery.